

Linux 기본 명령어 및 사용 가이드

2026 한국연소학회 가을 Tutorial
Next-Gen Combustion

Linux command

- Linux basic command*

Linux command	Description	Example
mkdir	Create a directory	mkdir directory
ls	List contents of a directory	ls /path/directory1
cd	Change directory with a specific path	cd /path/directory1
pwd	Print working directory	pwd
touch	create, change, and modify timestamps of a file	1. touch 2. touch file1
clear	Clear the screen	clear
cp	Copy file (s)	cp /path1/file1 /path2/file1
grep	Find a string of text in a file	grep 'word or phrase' file1
mv	Move file(s) or rename file(s)	mv /path1/file1 /path2/file2
rm	delete file(s)	rm file1
vi	Edit file(s) with simple text editor	vi file1

Linux command

● Linux basic command – **mkdir** & **ls**

Command Line

```
mkdir test
```

```
ls
```

Make Directory **list**

```
[cfd04@cheetah ~]$ mkdir test
[cfd04@cheetah ~]$ ls
test
```

test directory is generated

Command Line

```
ls -l (or ll)
```

-l option: list one file per line

```
[cfd04@cheetah ~]$ ls -l
total 4
drwxr-xr-x 2 cfd04 cfdgroup 4096 Mar  9 10:45 test
```

Command Line

```
ls --help
```

```
[cfd04@cheetah ~]$ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILES (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.
-a, --all                do not ignore entries starting with .
-A, --almost-all        do not list implied . and ..
--author                 with -l, print the author of each file
-b, --escape              print C-style escapes for nongraphic characters
--block-size=SIZE        scale sizes by SIZE before printing them; e.g.,
                          '--block-size=M' prints sizes in units of
                          1,048,576 bytes; see SIZE format below
-B, --ignore-backups     do not list implied entries ending with ~
-c                        with -lt: sort by, and show, ctime (time of last
                          modification of file status information);
                          with -l: show ctime and sort by name;
                          otherwise: sort by ctime, newest first
-C                        list entries by columns
--color[=WHEN]           colorize the output; WHEN can be 'never', 'auto',
                          or 'always' (the default); more info below
-d, --directory          list directories themselves, not their contents
-D, --dired               generate output designed for Emacs' dired mode
-f                        do not sort, enable -aU, disable -ls --color
-F, --classify            append indicator (one of */=>@|) to entries
                          likewise, except do not append '*'
--file-type              across -x, commas -m, horizontal -x, long -l,
                          single-column -l, verbose -l, vertical -C
--format=WORD             like -l --time-style=full-iso
--full-time              like -l, but do not list owner
-g                        like -l, but do not list owner
```

Linux command

- Linux basic command – **cd** & **pwd**

Command Line

pwd

```
[cfd04@cheetah ~]$ pwd  
/uhome/edcfd/cfd04
```

print **w**orking **d**irectory

Command Line

cd test

pwd

```
[cfd04@cheetah ~]$ cd test  
[cfd04@cheetah test]$ pwd  
/uhome/edcfd/cfd04/test
```

change **d**irectory to *test* directory

- Linux basic command – **touch**

Command Line

touch a

ls -l

```
[cfd04@cheetah test]$ touch a  
[cfd04@cheetah test]$ ls -l  
total 0  
-rw-r--r-- 1 cfd04 cfdgroup 0 Mar  9 10:56 a
```

black file *a* is
generated

Linux command

● Linux basic command – **cp**

Command Line

```
cp a b
```

```
ls -l
```

```
[cfd04@cheetah test]$ cp a b
[cfd04@cheetah test]$ ls -l
total 0
-rw-r--r-- 1 cfd04 cfdgroup 0 Mar  9 10:56 a
-rw-r--r-- 1 cfd04 cfdgroup 0 Mar  9 10:58 b
```

→ **copy a to b**

● Linux basic command – **cd ..**

- .. : parent directory (상위 디렉토리)
- . : current directory (현재 디렉토리)

Command Line

```
cd ../
```

```
pwd
```

```
ls -l
```

```
[cfd04@cheetah test]$ cd ../
[cfd04@cheetah ~]$ pwd
/uhome/edcfd/cfd04
[cfd04@cheetah ~]$ ls -l
total 4
drwxr-xr-x 2 cfd04 cfdgroup 4096 Mar  9 10:58 test
```

→ **Change directory to the parent directory**

Linux command

● Linux basic command – **rm**

Command Line

```
rm a
```

```
ls -l
```

```
[cfd04@cheetah test3]$ rm a  
[cfd04@cheetah test3]$ ls -l  
total 4  
-rw-r--r-- 1 cfd04 cfdgroup  0 Mar  9 11:07 b  
drwxr-xr-x 2 cfd04 cfdgroup 4096 Mar  9 10:58 test
```

remove file a

● Linux basic command – **rm -options**

```
[cfd04@cheetah test3]$ rm test  
rm: cannot remove 'test': Is a directory
```

rm command basically remove **file** such that directory is not allowed to be copied

Command Line

```
rm -r test
```

```
ll
```

```
[cfd04@cheetah test3]$ rm -r test  
[cfd04@cheetah test3]$ ll  
total 0  
-rw-r--r-- 1 cfd04 cfdgroup 0 Mar  9 11:07 b
```

-r option allows to remove directory

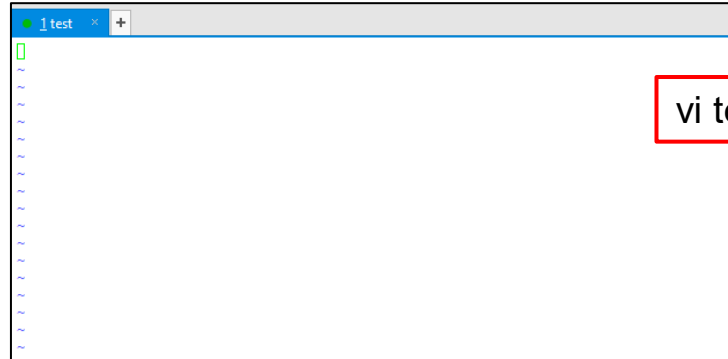
```
[cfd04@cheetah test3]$ rm --help  
Usage: rm [OPTION]... FILE...  
Remove (unlink) the FILE(s).  
  
-f, --force          ignore nonexistent files and arguments, never prompt  
-i                   prompt before every removal  
-I                   prompt once before removing more than three files, or  
                    when removing recursively; less intrusive than -i,  
                    while still giving protection against most mistakes  
--interactive[=WHEN] prompt according to WHEN: never, once (-I), or  
                    always (-i); without WHEN, prompt always  
--one-file-system    when removing a hierarchy recursively, skip any  
                    directory that is on a file system different from  
                    that of the corresponding command line argument  
--no-preserve-root   do not treat '/' specially  
--preserve-root      do not remove '/' (default)  
-r, -R, --recursive remove directories and their contents recursively  
-d, --dir            remove empty directories  
-v, --verbose        explain what is being done  
--help              display this help and exit  
--version            output version information and exit
```

Linux command

- Utilizing **vi** editor

Command Line

```
vi b
```



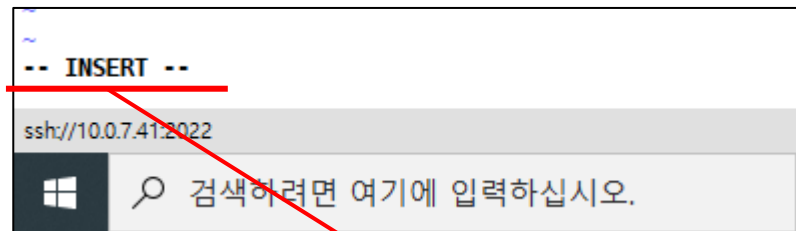
vi test editor is working for file *b*

vi command can open existing file, or make new file

Keyboard

press "i"

Or press "insert"

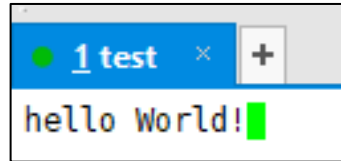


"insert" option is activated. **Insert mode** (See the lower-left corner)

Linux command

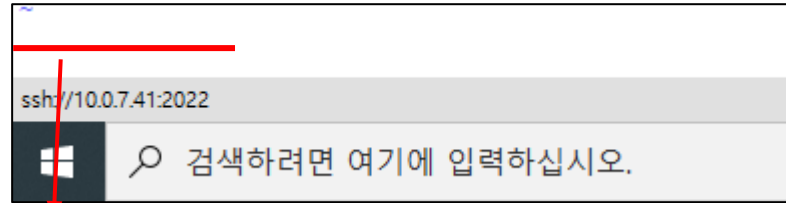
● Utilizing **vi** editor

Type in vi file
(Insert mode)



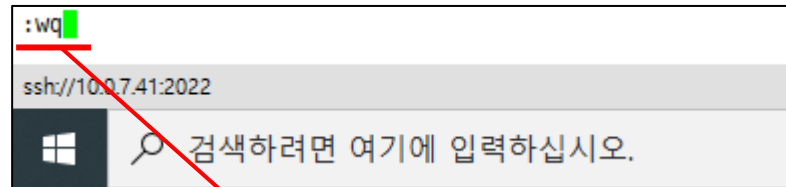
Typing is available in **insert mode**.

Press "Esc" key



"insert" option is de-activated. **Command mode**

Type :wq
(command mode)
Press enter key



:wq write and quit the file
:w write
:q quit the file
:q! quit the file (ignore the change)